Volume of Prisms

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

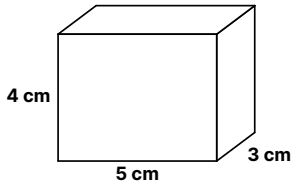
KEY WORDS volume *(rōrahi)* · prism *(poro)* · cube *(mataono)* · rectangular prism *(poro tapawhā hāngai)*

A Rectangular prisms ●○○

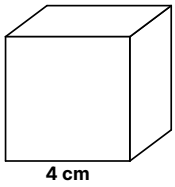
Remember: Volume of a rectangular prism = length × width × height. Volume is measured in cubed units, such as cm^3.

1. Find the volume of each solid.

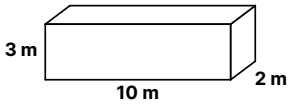
a)



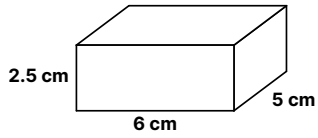
b)



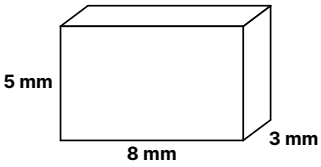
c)



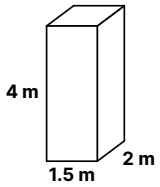
d)



e)



f)



2. Find the volume of:

a) A rectangular prism 12 cm by 5 cm by 2 cm

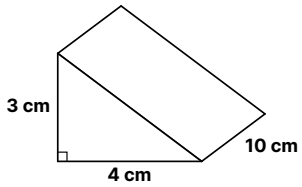
b) A cube with edges of 3 m

B Triangular prisms ●●○

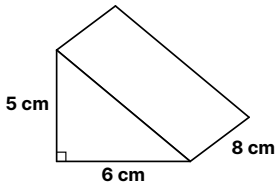
Remember: Volume of a prism = area of the end × length. For a triangular prism the end is a triangle, so $V = \frac{1}{2} \times b \times h \times l$.

3. Find the volume of each triangular prism.

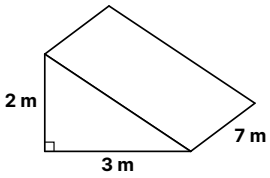
a)



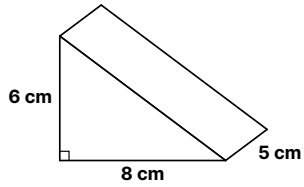
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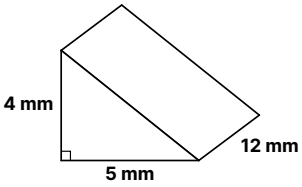
c)



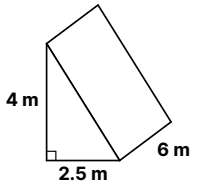
d)



e)



f)



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4. Find the volume of:

a) A triangular prism whose end is a triangle with base 10 cm and height 6 cm, and whose length is 4 cm

b) A triangular prism whose end is a triangle with base 3 m and height 1.2 m, and whose length is 5 m

C Composite solids ●●●

Remember: Split the solid into prisms, find the volume of each one, then add the volumes.

5. Find the volume of each solid.

a) A rectangular prism 6 cm by 4 cm by 3 cm with a cube of side 2 cm glued on top

b) A shed made from a rectangular prism 10 m long, 6 m wide and 4 m high, with a roof that is a triangular prism. The triangle at the end of the roof has a base of 6 m and a height of 3 m, and the roof is 10 m long.

c) A step made from two rectangular prisms: one 10 cm by 4 cm by 2 cm, and one 5 cm by 4 cm by 2 cm on top of it

D Missing dimensions ●●●

Remember: Work backwards. If volume = area of the end × length, then length = volume ÷ area of the end.

6. Find the missing length.

a) A rectangular prism has a volume of 120 cm^3. It is 6 cm long and 5 cm wide. Find its height.

b) A prism has a volume of 90 cm^3. The area of its end is 15 cm^2. Find its length.

c) A cube has a volume of 64 cm^3. Find the length of its edges.

d) A rectangular prism has a volume of 84 m^3. It is 7 m long and 4 m wide. Find its height.

e) A triangular prism has a volume of 60 cm^3. Its end is a triangle with base 4 cm and height 3 cm. Find its length.

f) A prism has a volume of 31.5 m^3. The area of its end is 9 m^2. Find its length.

E Problems ●●●

7. A fish tank is 50 cm long, 30 cm wide and 20 cm deep. How many litres of water does it hold when it is full? (1 litre = 1,000 cm^3)

8. A sandpit is 2 m long and 1.5 m wide. Sand is spread in it to a depth of 0.4 m. What volume of sand is needed?

★ Extension

9. A cube has edges of 3 cm. Every edge is doubled. How many times larger is the volume of the new cube?

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10. Small boxes are 4 cm by 3 cm by 2 cm. A carton with no gaps holds exactly 48 of them. What is the volume of the carton?

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Volume of Prisms

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

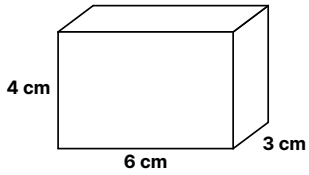
KEY WORDS volume *(rōrahi)* · prism *(poro)* · cube *(mataono)* · rectangular prism *(poro tapawhā hāngai)*

A Rectangular prisms ●○○

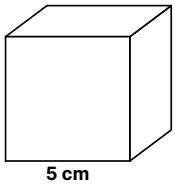
Remember: Volume of a rectangular prism = length × width × height. Volume is measured in cubed units, such as cm^3.

1. Find the volume of each solid.

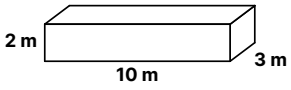
a)



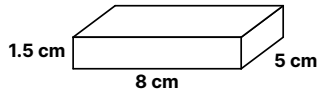
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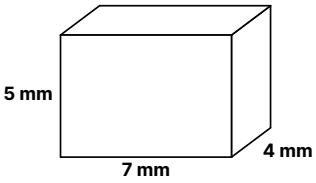
c)



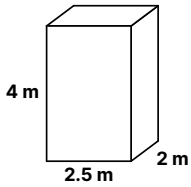
d)



e)



f)



2. Find the volume of:

a) A rectangular prism 15 cm by 4 cm by 2 cm

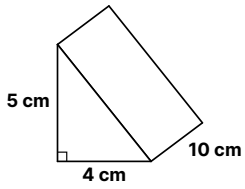
b) A cube with edges of 2 m

B Triangular prisms ●●○

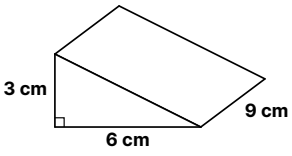
Remember: Volume of a prism = area of the end × length. For a triangular prism the end is a triangle, so $V = \frac{1}{2} \times b \times h \times l$.

3. Find the volume of each triangular prism.

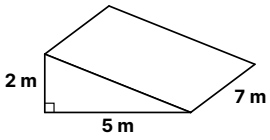
a)



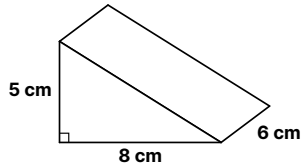
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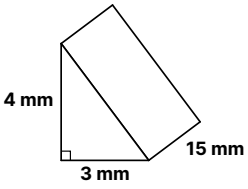
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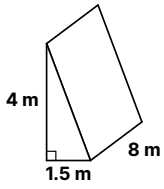
d)



e)



f)



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4. Find the volume of:

a) A triangular prism whose end is a triangle with base 12 cm and height 5 cm, and whose length is 4 cm

b) A triangular prism whose end is a triangle with base 4 m and height 1.5 m, and whose length is 5 m

C Composite solids ●●●

Remember: Split the solid into prisms, find the volume of each one, then add the volumes.

5. Find the volume of each solid.

a) A rectangular prism 8 cm by 5 cm by 2 cm with a cube of side 3 cm glued on top

b) A shed made from a rectangular prism 12 m long, 5 m wide and 4 m high, with a roof that is a triangular prism. The triangle at the end of the roof has a base of 5 m and a height of 3 m, and the roof is 12 m long.

c) A step made from two rectangular prisms: one 8 cm by 5 cm by 3 cm, and one 4 cm by 5 cm by 3 cm on top of it

D Missing dimensions ●●●

Remember: Work backwards. If volume = area of the end × length, then length = volume ÷ area of the end.

6. Find the missing length.

a) A rectangular prism has a volume of 160 cm^3. It is 8 cm long and 5 cm wide. Find its height.

b) A prism has a volume of 96 cm^3. The area of its end is 12 cm^2. Find its length.

c) A cube has a volume of 125 cm^3. Find the length of its edges.

d) A rectangular prism has a volume of 72 m^3. It is 6 m long and 4 m wide. Find its height.

e) A triangular prism has a volume of 80 cm^3. Its end is a triangle with base 5 cm and height 4 cm. Find its length.

f) A prism has a volume of 38.5 m^3. The area of its end is 11 m^2. Find its length.

E Problems ●●●

7. A fish tank is 60 cm long, 25 cm wide and 20 cm deep. How many litres of water does it hold when it is full? (1 litre = 1,000 cm^3)

8. A sandpit is 3 m long and 1.5 m wide. Sand is spread in it to a depth of 0.2 m. What volume of sand is needed?

★ Extension

9. A cube has edges of 2 cm. Every edge is doubled. How many times larger is the volume of the new cube?

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10. Small boxes are 5 cm by 2 cm by 2 cm. A carton with no gaps holds exactly 60 of them. What is the volume of the carton?

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